and stronger—he has to hold the cable tighter and tighter as he gets closer to the sun to avoid being pulled in. Each of them agrees that George's clock is running slow. There is no "equally valid perspective" that exchanges their roles and reverses this conclusion. This is, in fact, what we found in Chapter 2 when George experienced an acceleration by turning on his jetpack to catch up with Gracie. The acceleration George felt resulted in his clock definitively running slow relative to Gracie's. Since we now know that feeling accelerated motion is the same as feeling a gravitational force, the present situation of George on the cable involves the same principle, and once again we see that George's clock, and everything else in his life, runs in slow motion compared with Gracie's.

In a gravitational field such as that at the surface of an ordinary star like the sun, the slowing of clocks is quite small. If Gracie stays put at a billion miles from the sun, then when George has climbed to within a few miles of its surface, the rate of ticking of his clock will be about 99.9998 percent of Gracie's. Slower, but not by much.9 If, however, George lowered himself on a cable so that he hovered just above the surface of a neutron star whose mass, roughly equal to that of the sun, is crushed to a density some million billion times that of solar density, the larger gravitational field would cause his clock to tick at about 76 percent of the rate of Gracie's. Stronger gravitational fields, such as those just outside a black hole (as discussed below), cause the flow of time to slow even further; stronger gravitational fields cause a more severe warping of time.

Experimental Verification of General Relativity

Most people who study general relativity are captivated by its aesthetic elegance. By replacing the cold, mechanistic Newtonian view of space, time, and gravity with a dynamic and geometric description involving curved spacetime, Einstein wove gravity into the basic fabric of the universe. Rather than being imposed as an additional structure, gravity becomes part and parcel of the universe at its most fundamental level. Breathing life into space and time by allowing them to curve, warp, and ripple results in what we commonly refer to as gravity.

Aesthetics aside, the ultimate test of a physical theory is its ability to explain and predict physical phenomena accurately. Since its